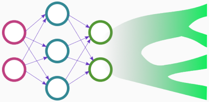
Likelihood-free phylogenetics



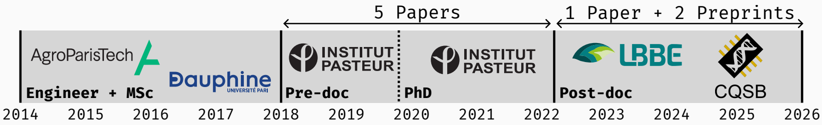
Luc Blassel

Auditions CNRS 51/02

May 20th 2026

slides: `lucblassel.com/CNRS_2026.pdf`

Personal research context & history



An interdisciplinary setting:

- *Algorithmics & theory*: **Improving** sequence **alignments**
- *Data analysis*: **Learning** from alignments
- *Methods & engineering*: **Deep Learning** for phylogenetic **inference**

General research framework:

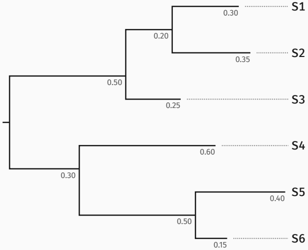
Methods to gather **evolutionary** insight from **sequences**

```
S1 PRLPTC
S2 TQPSVPC
S3 TGVPVC
S4 TLRSPC
S5 TQPRKTC
S6 TQPAC
```

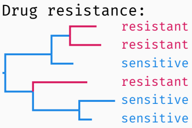
Sequences

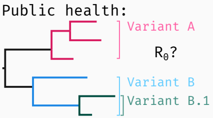
```
S1 --PRLPTC
S2 TQPSVP-C
S3 TGVPV--C
S4 TL-RSP-C
S5 TQPR-KTC
S6 TQ---PAC
```

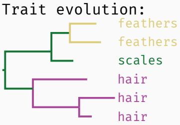
Alignment



Tree

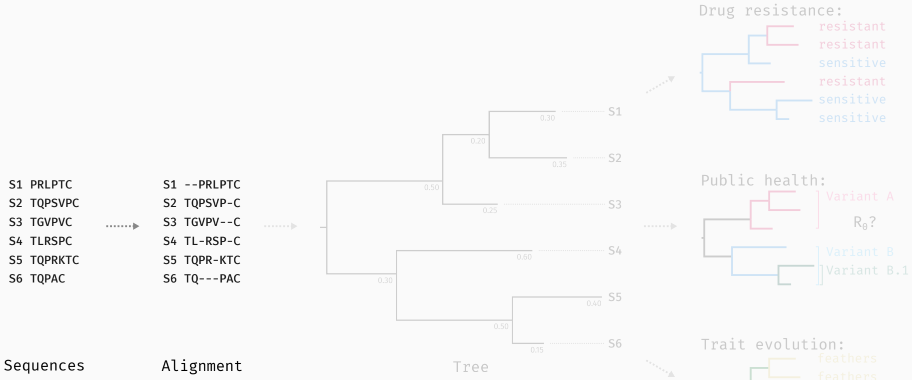






Previous research work

keywords: Sequence Bioinformatics, combinatorics



Blassel *et al.* 2022, RECOMB-SEQ;

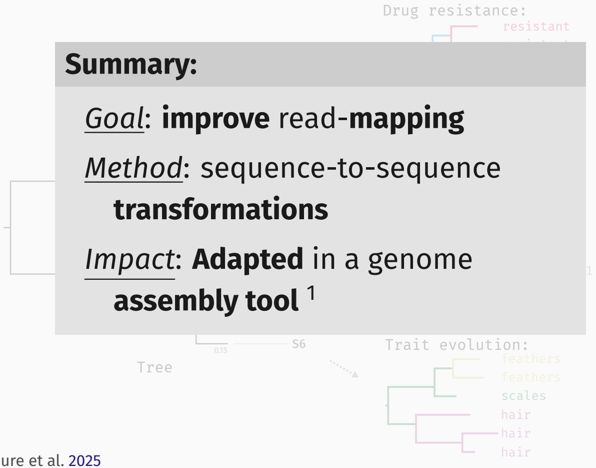
keywords: Sequence Bioinformatics, combinatorics

```
S1 PRLPTC
S2 TQPSVPC
S3 TGVPVC
S4 TLRSPC
S5 TQPRKTC
S6 TQPAC
```

Sequences

```
S1 --PRLPTC
S2 TQPSVP-C
S3 TGVPV--C
S4 TL-RSP-C
S5 TQPR-KTC
S6 TQ---PAC
```

Alignment



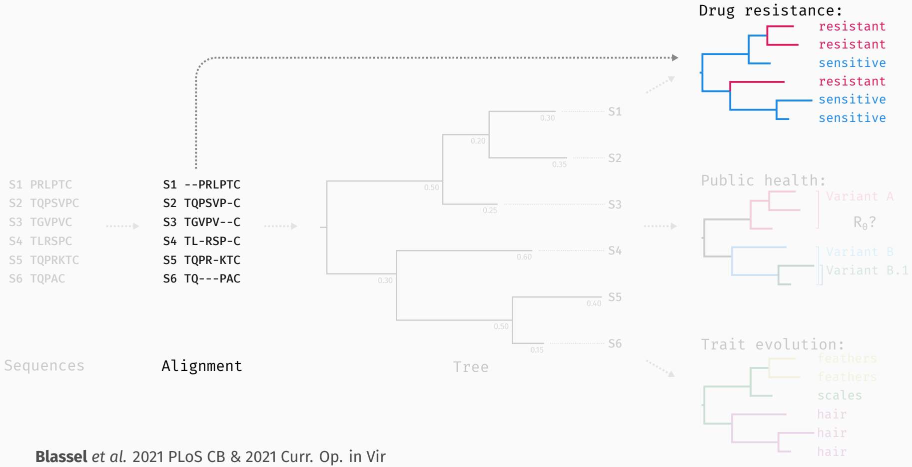
Summary:

Goal: **improve** read-**mapping**

Method: sequence-to-sequence **transformations**

Impact: **Adapted** in a genome **assembly tool** [1]

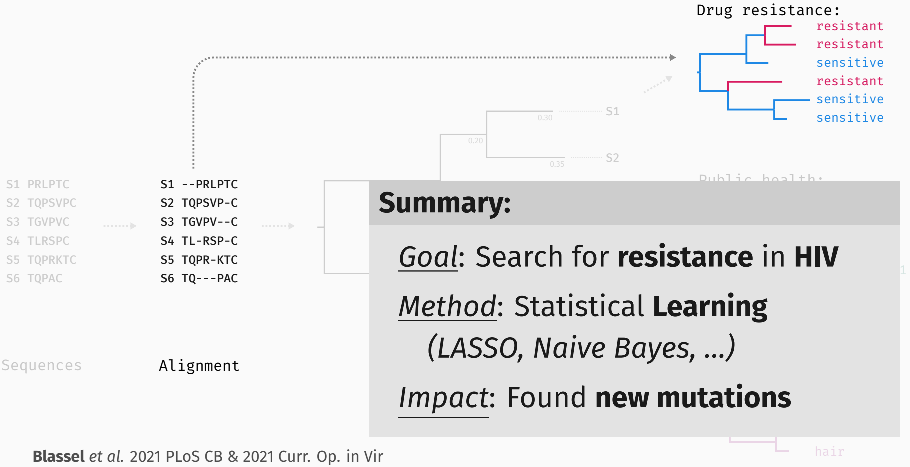
Blassel *et al.* 2022, RECOMB-SEQ; [1]Faure et al. 2025 *keywords*: Data Analysis, Pipelining, Machine Learning



Blassel *et al.* 2021 PLoS CB & 2021 Curr. Op. in Vir

Previous works - Learning from alignments

keywords: Data Analysis, Pipelining, Machine Learning



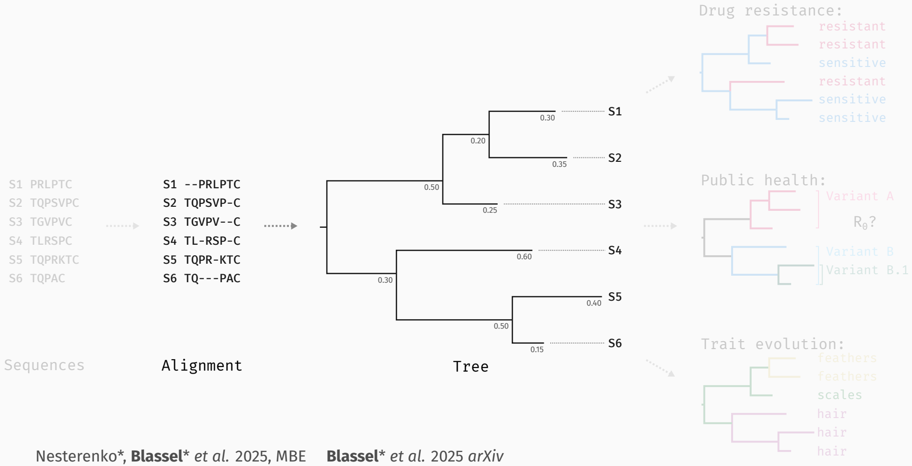
Summary:

Goal: Search for **resistance** in **HIV**

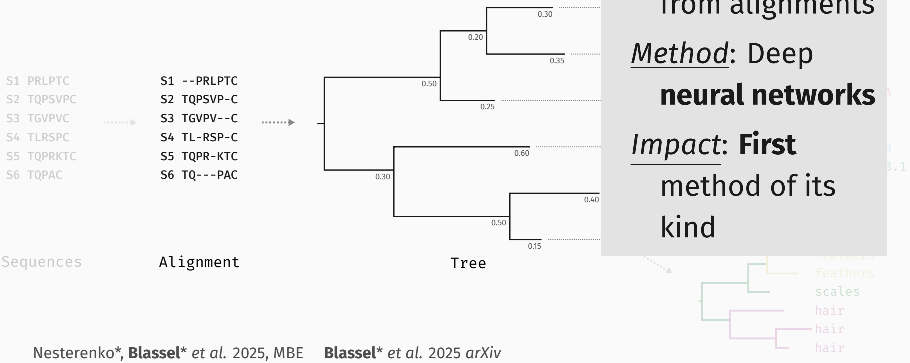
Method: Statistical **Learning** (*LASSO, Naive Bayes, ...*)

Impact: Found **new mutations**

Blassel *et al.* 2021 PLoS CB & 2021 Curr. Op. in Vir *keywords*: Pytorch, Deep Learning,



Nesterenko*, **Blassel*** *et al.* 2025, MBE **Blassel*** *et al.* 2025 *arXiv* *keywords*: Pytorch, Deep Learning,



Summary:

Goal: **Infer trees** from alignments

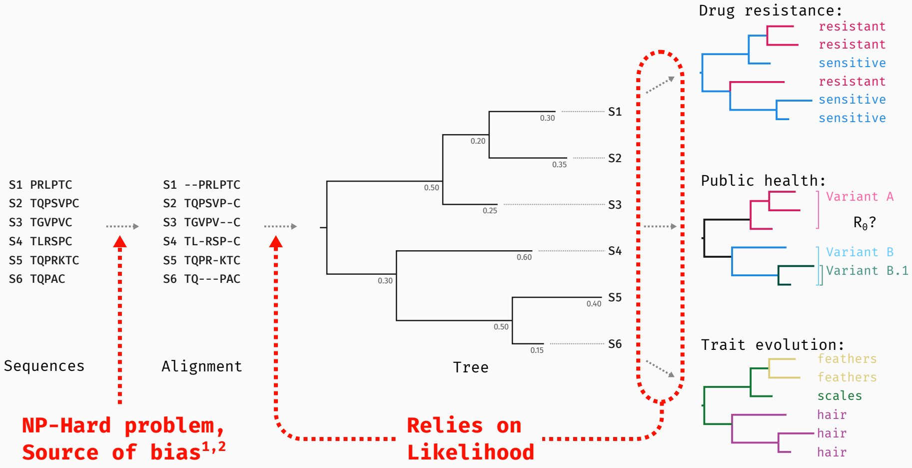
Method: Deep **neural networks**

Impact: **First** method of its kind

Nesterenko*, **Blassel*** *et al.* 2025, MBE **Blassel*** *et al.* 2025 *arXiv*

Context - Methodological Barriers

[1]Liu et al. 2011,[2]Wong et al. 2008



Context - The issue with likelihoods

State of the art relies on **Likelihood ($\mathcal{L}$)**:

Max. Likelihood: $\theta^* = \underset{\theta}{\operatorname{argmax}}\ \mathbf{p}(\mathbf{x}|\theta, \mathbf{M})$

Bayesian: $p(\theta|x, M) \propto \mathbf{p}(\mathbf{x}|\theta, \mathbf{M}) p(\theta) p(M)$

Costly[1,2]

$\mathcal{L} = \mathcal{O}(ns^3 + nLs^2)$

Limits model-complexity

Settings with **impractical** $\mathcal{L}$

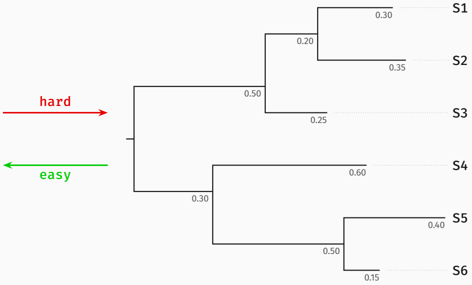
- **Potts**: models site **co-evolution**[3]
- **MTBD**: can model **epidemics** with distinct **populations**[4]

n: nb. of leaves, L: length of sequence, s: size of alphabet

[1]Bryant et al. 2005, [2]Felsenstein 2004, [3]Rodriguez Horta et al. 2019, [4]Barido-Sottani et al. 2020

Context - Likelihood-free inference

S1 TQPRLPTC
S2 TQPSVPKC
S3 TGVPVPAC
S4 TLGRSPSC
S5 TQPRAKTC
S6 TQPRIPAC

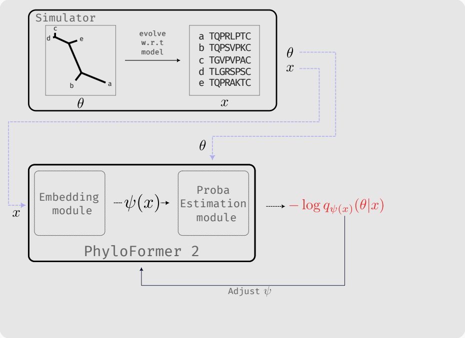


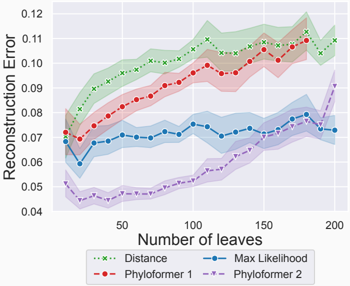
Setting:

- **Hard** to compute **likelihood**
- **Easy** to **simulate**

Solution:

1. **Simulate** many datapoints **under model priors**
2. **Learn** $f(x) : x \mapsto \theta$





Goal: $q_{\psi(x)}(\theta|x) \approx p(\theta|x)$

Training: $\psi^* = \text{argmin}_\psi \; -\sum_{i \in train} \log q_{\psi(x_i)}(\theta_i|x_i)$ [1]

Inference: we **sample** from $q_{\psi^*(x_E)}(\theta_E|x_E)$

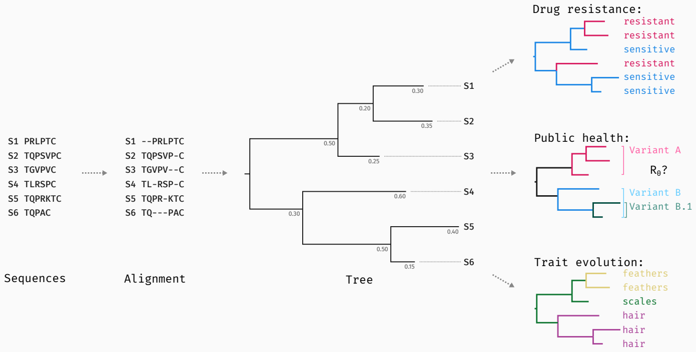
Blassel *et al.* 2025, *preprint*

[1] equiv. to: $\text{argmin}_\psi \; D_{KL}\left(q_{\psi(x)}(\theta|x)||p(\theta|x)\right)$

Research Project

Goal:

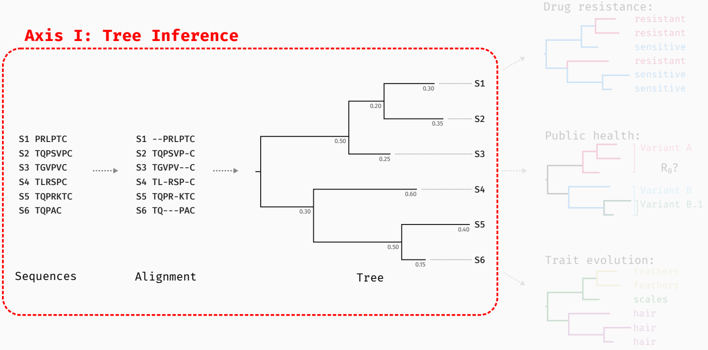
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Overview

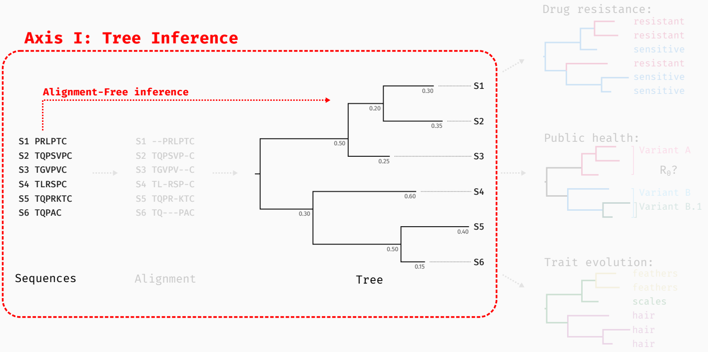
Goal:

likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



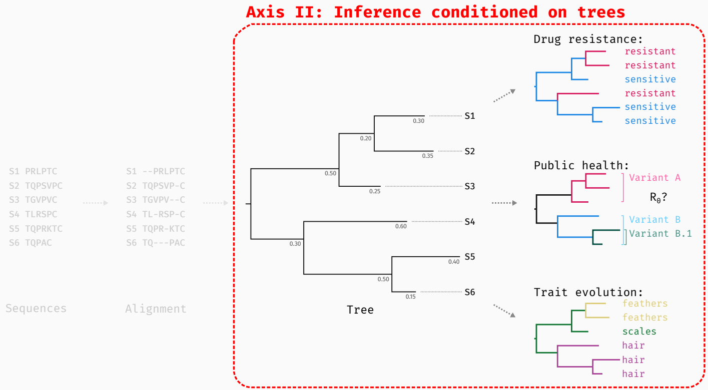
Goal:

likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



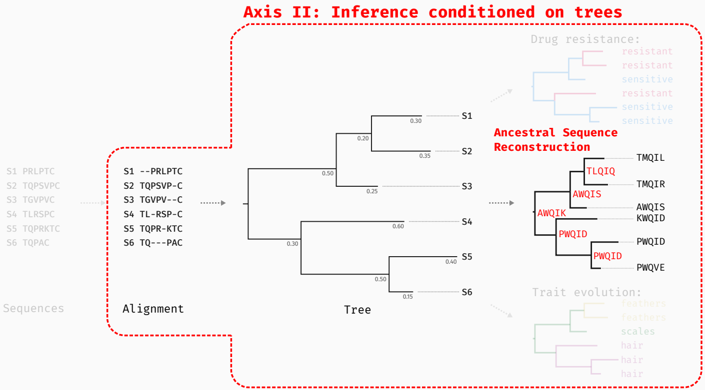
Goal:

likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



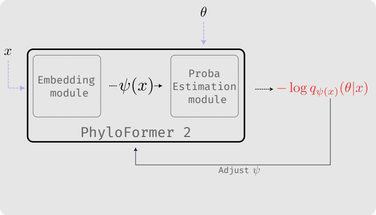
Goal:

likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Problem (Reminder):

- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can be an important source of bias[1,2]



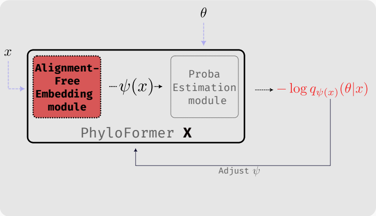
- **Adapt** our existing posterior estimation **framework**

[1] Liu et al. 2011, [2] Wong et al. 2008

Project - Ax. I - Alignment and likelihood-free inference

Problem (Reminder):

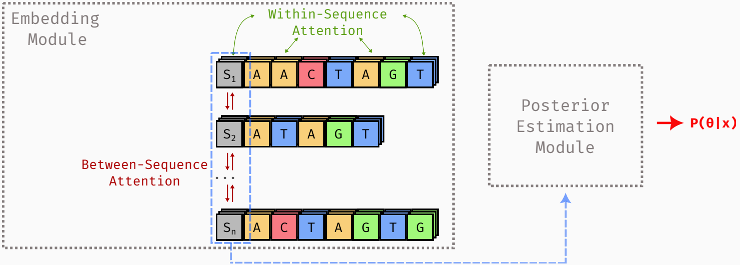
- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can be an important source of bias[1,2]



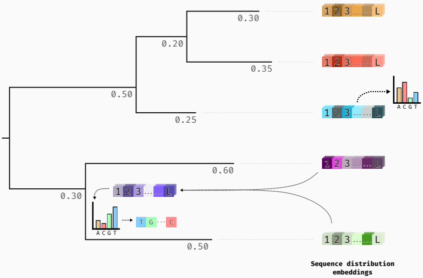
- **Adapt** our existing posterior estimation **framework**
- Develop a way of **representing unaligned sequences**

[1] Liu et al. 2011, [2] Wong et al. 2008

Project - Ax. I - Alignment and likelihood-free inference



- **Re-use** existing posterior estimation **module**
- **Easy** to create a first **prototype**
- *Challenges*: overcome **information bottlenecks** & adaptation to **empirical data**



- **per-site distribution** over alphabet
- *Training*: we have **simulated ancestral** seq.
- **Flexible** approach for other **branch-specific** parameters: Dates, speciation rates, . . .

Project - Next steps (short-medium term)

PF2 presents **posterior calibration** issues

$\Rightarrow$ More **flexible posterior** modelling with **diffusion**

Alignments can still be **useful**

$\Rightarrow$ **Integrate** alignment-free approach **to MSA** pipeline

Can we model **more complex** evolutionary **processes**?

$\Rightarrow$ Infer **tree-sequences** to start tackling **recombination**

Project - Next Steps (long term)

Scaling to large number of sequences:

- **17M** SARS-CoV-2 genomes in GISAID
- **900k** genomes in GTDB
- **250k** genomes in IGM database

Extend from **trees** to **networks**:

- **Horizontal transfer** is common in **bacteria**[1]
- **Recombination** is frequent in **viruses**[2]
- *Collaboration with UW-Madison*

numbers: GISAID, GTDB, IMG

[1]Arnold et al. 2022, [2]Pérez-Losada et al. 2015

Integration & current collaborations

Analytical Genomics team @ **CQSB**:

Expertise: SBI/ML (L. Jacob), Evolution (M. Weigt, A. Carbone)

Collaborators: L. Jacob & P. Barrat-Charlaix, M. Weigt *(planned)*

Bioinfo team @ **LISN**:

Expertise: SBI/ML (F. Pouyet, F. Jay), Trees/Graphs (L. Planche)

Collaborators: F.Pouyet & F. Jay *(planned)*

Modelling of Pathogens team @ **IBENS** & **IP**:

Expertise: Evolution & Modelling (A. Zhukova, H. Morlon)

Collaborators: A. Zhukova

Research output:

6 Peer-reviewed **papers**
incl. ***4*** *as (co)1st author*
2 recent **Preprints**.
Code on `git(hub|lab)`.

Supervision:

L3: *1*, **M1**: *1*, **M2**: *2*
Oct 2026: **PhD** co-supervision with **A. Zhukova**

Reviewing:

Journals: MBE, Syst. Bio, PLoS CB, Bioinformatics
Conferences: RECOMB-SEQ, ICLR, ICML

Collaborations:

France: CIRB, IBENS, Institut Pasteur, LBBE
International: Penn. State, EPFL, UW-Madison

Funded by:

PhD: INCEPTION & PRAIRIE
PostDoc: ANR DEELOGENY

Thank you for listening!

References

Arnold, B. J. et al. (2022). ***Horizontal Gene Transfer and Adaptive Evolution in Bacteria*.** In: *Nature Reviews Microbiology* 20.4, pp. 206–218.

Barido-Sottani, J. et al. (2020). ***A Multitype Birth–Death Model for Bayesian Inference of Lineage-Specific Birth and Death Rates*.** In: *Syst Biol* 69.5, pp. 973–986.

Bryant, D. et al. (2005). ***Likelihood Calculation in Molecular Phylogenetics*.** In: *Mathematics of Evolution and Phylogeny*. Ed. by O. Gascuel. Oxford University PressOxford, pp. 33–62.

Faure, R. et al. (2025). ***Alice: Fast and Haplotype-Aware Assembly of High-Fidelity Reads Based on MSR Sketching*.** Pre-published.

Felsenstein, J. (2004). ***Inferring Phylogenies*.**

Liu, K. et al. (2011). ***Multiple Sequence Alignment: A Major Challenge to Large-Scale Phylogenetics*.** In: *PLoS Curr* 2, RRN1198.

Pérez-Losada, M. et al. (2015). ***Recombination in Viruses: Mechanisms, Methods of Study, and Evolutionary Consequences*.** In: *Infect Genet Evol* 30, pp. 296–307.

Rodriguez Horta, E. et al. (2019). ***Toward Inferring Potts Models for Phylogenetically Correlated Sequence Data*.** In: *Entropy* 21.11, p. 1090.

Wong, K. M. et al. (2008). ***Alignment Uncertainty and Genomic Analysis*.** In: *Science* 319.5862, pp. 473–476.

Supplementary slides

Project - Assembly-free phylogenetic inference

Schema

Project - Tree-sequence inference

frame

Project - Inference on trees, GNNs for R_0

frame

Project - Joint tree-parameter inference, dating

env